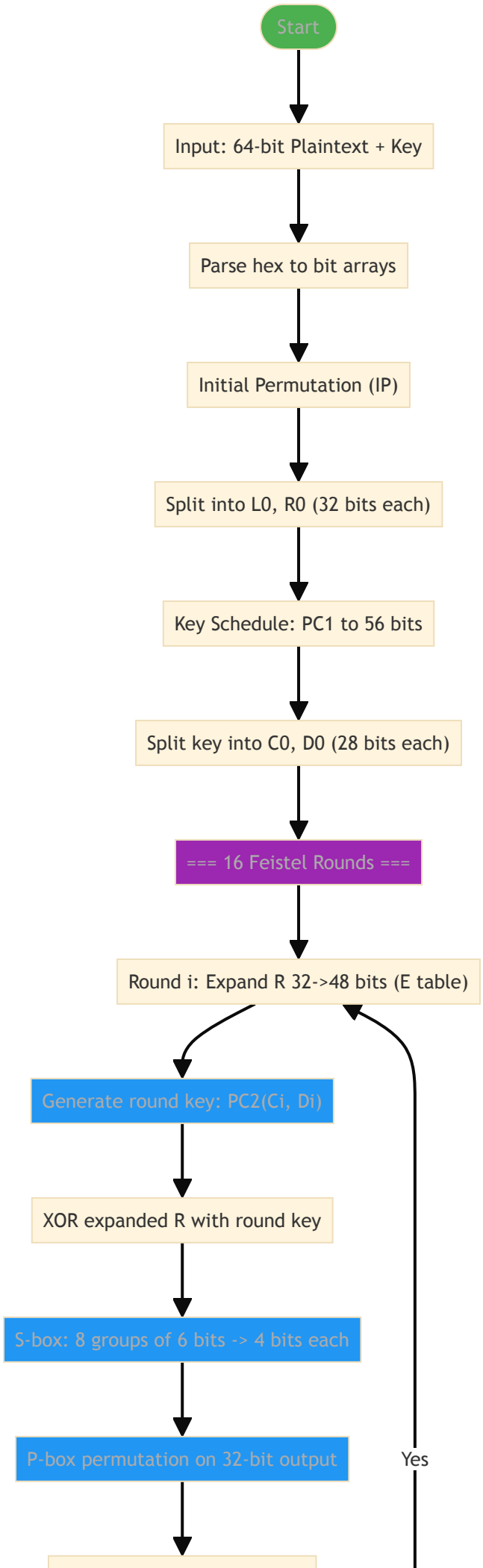
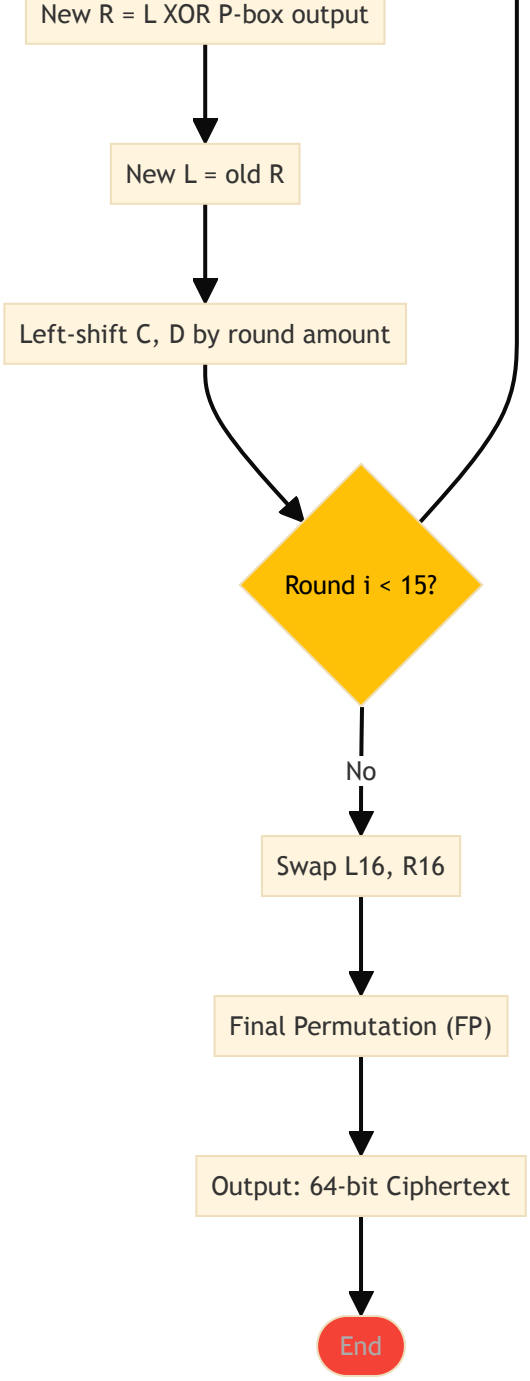

Start
Input: 64-bit Plaintext + Key
Parse hex to bit arrays
Initial Permutation (IP)
Split into L0, R0 (32 bits each)
Key Schedule: PC1 to 56 bits
Split key into C0, D0 (28 bits each)
=== 16 Feistel Rounds ===
Round i: Expand R 32->48 bits (E table)
Generate round key: PC2(Ci, Di)
XOR expanded R with round key
S-box: 8 groups of 6 bits -> 4 bits each
P-box permutation on 32-bit output
Yes


New R = L XOR P-box output
New L = old R
Left-shift C, D by round amount
Round i < 15?
No
Swap L16, R16
Final Permutation (FP)
Output: 64-bit Ciphertext
End